JAVA-BASED ENTERPRISE INVENTORY MANAGEMENT SYSTEM

Project Code: #z-13-18
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Batch- 13

INTRODUCTION

Inventory management is essential for tracking stock and ensuring smooth business operations.

Manual systems lead to:

- Errors in stock calculation
- Delayed updates
- Lack of insights

This project presents a Java-based enterprise inventory management system using:

- Spring Boot for backend
- Thymeleaf for dynamic UI

The system enables:

- Automated stock tracking
- Secure access
- Data visualization and reporting

PROBLEM STATEMENT

Existing inventory systems lack:

- Automation
- Real-time updates
- Secure user management

Businesses face:

- Overstocking and understocking
- Inefficient tracking
- Poor decision-making due to lack of analytics

The solution needed:

A need exists for a **centralized, secure, and intelligent inventory system**.

PROJECT OBJECTIVES

01

Build a web-based inventory system using Spring Boot.

02

Implement secure authentication and authorization using Spring Security.

03

Enable real-time stock updates and monitoring.

04

Provide visual insights using charts (Chart.js).

05

Generate reports and export data using OpenCSV.

06

Send email alerts for low stock levels using JavaMail API.

SYSTEM ARCHITECTURE

Architecture Type: MVC (Model-View-Controller)

View Layer
- Thymeleaf templates
- HTML, CSS, Bootstrap

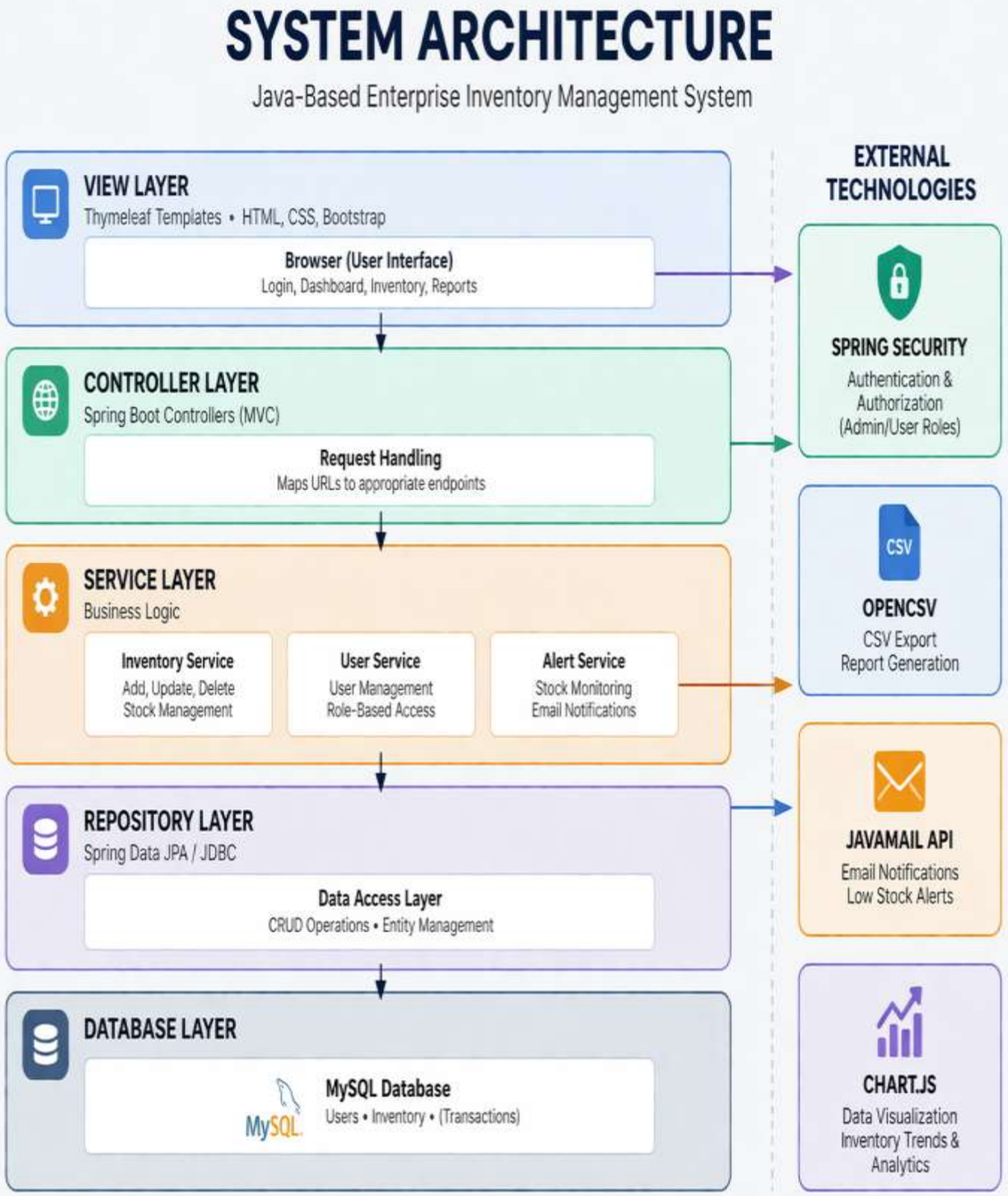
Controller Layer
- Spring Boot Controllers
- Handles HTTP requests

Service Layer
- Business logic (inventory handling, alerts)

Repository Layer
- JPA / JDBC interaction with MySQL

Security Layer
- Spring Security (authentication & authorization)

Database Layer
- MySQL



TECHNOLOGY STACK

Layer	Technology
Frontend	HTML, CSS, Bootstrap, Thymeleaf
Backend	Java, Spring Boot
Database	MySQL
Security	Spring Security
Visualization	Chart.js
Tools	IntelliJ IDEA, Git

SYSTEM MODULES

1. Inventory Tracking Engine

- Add, update, delete inventory items
- Tracks stock levels in real time
- Stores data in MySQL

2. User Authentication System

- Secure login using Spring Security
- Role-based access: Admin User

3. Reporting & CSV Integration Hub

- Generates inventory reports
- Exports data using OpenCSV
- Enables external data usage

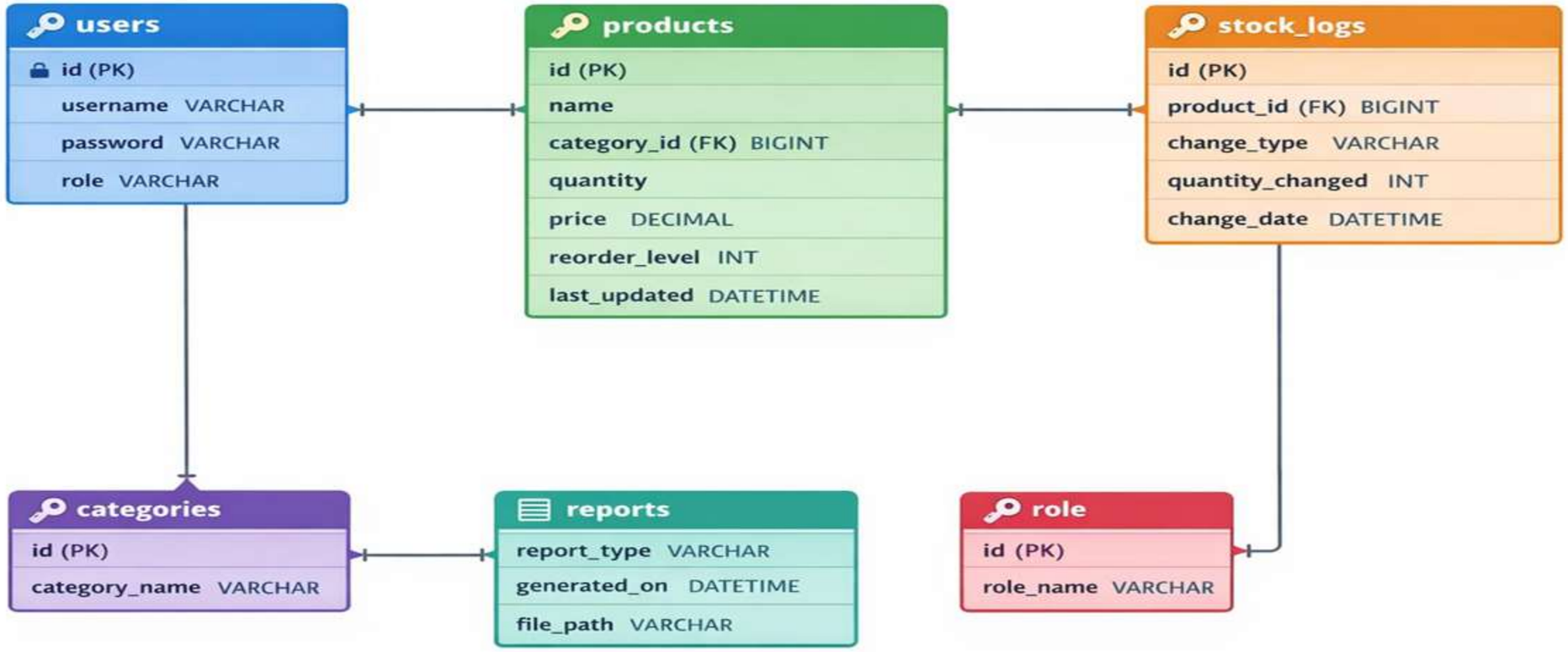
4. Stock Alert & Management Coach

- Monitors stock levels continuously
- Sends email alerts using JavaMail API
- Helps maintain optimal inventory

5. Data Visualization Module

- Displays inventory trends using Chart.js
- Helps in decision-making

DATABASE DESIGN



KEY FEATURES

- Secure authentication using Spring Security
- Role-based access control
- Real-time inventory tracking
- Email notifications for low stock
- CSV report export
- Interactive charts using Chart.js
- User-friendly web interface

CONCLUSION

Project Success

Successfully developed a **secure and efficient inventory system**.

Technical Excellence

The system demonstrates:

- Strong backend design using Spring Boot
- Integration of multiple technologies

Key outcomes:

- Reduced manual effort
- Improved accuracy
- Better stock visibility

Future extensions:

- Multi-user systems
- Cloud deployment

THANK YOU